A Spectral–Spatial Adaptive Weighted Method for Shadow Removal in Vegetation Canopy Using Multiscale UAV-Based Hyperspectral Imagery

Wenxuan Zhang, Kangning Li, Jinbao Jiang, Ming Wang, Di Pan, Xiaojun Qiao, Xuepeng Xiao, and Xiaotong Ban

***Abstract*—While uncrewed aerial vehicles (UAVs) provide high spectral–spatial resolution and operational flexibility for hyperspectral vegetation monitoring, their high-resolution imagery remains susceptible to shadow interference, which obscures true canopy reflectance, thereby compromising classification and biophysical parameter retrieval accuracy. However, current shadow removal methods exhibit limited dynamic adaptability to varying shadow characteristics and demonstrate inconsistent performance across different spatial scales. To examine resolution effects, shadow characteristics were analyzed before shadow removal: with decreasing resolution, the absolute spectral difference between sunlit and shadowed regions diminished, the shadow area decreased, and brightness variation along shadow boundaries increased. Based on these observations, this study proposes a spectral–spatial adaptive weighted method (SAWM) for shadow removal for multiscale UAV-based hyperspectral imagery, which consists of three steps: 1) sunlit regions were preliminarily detected using two vegetation indices and the threshold-based approach; 2) endmembers were extracted from the sunlit regions using unsupervised algorithms and merged through K-means clustering; and 3) a joint optimization framework for shadow removal was constructed by integrating the shadow multilinear mixing (SMLM) model with total variation (TV) spatial regularization, with the regularization weight adaptively determined based on the two-indicator balance points. The results show that SAWM achieved a mean root-mean-square error (RMSE) of 0.0768 and relative RMSE (rRMSE) of 44.09% across multiple spatial resolutions. Compared to three representative methods, SAWM achieved improvements of up to 0.0332 in RMSE and 15.94% in rRMSE. SAWM offers an adaptive solution for shadow removal in high-resolution multiscale UAV imagery, with potential to improve vegetation analysis in complex environmental conditions.**

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***Index Terms*—Hyperspectral imagery, multiscale, shadow removal, uncrewed aerial vehicle (UAV), vegetation canopy.**

I. Introduction

As A fundamental component of terrestrial ecosystems, vegetation plays a crucial role in regulating the exchange of water, energy, and carbon fluxes between the land surface and the atmosphere [1]. Agricultural crops, as typical human-managed vegetation, contribute significantly to regional carbon sequestration and hydrological processes [2], [3]. The canopy characteristics of vegetation, including their structural and physiological traits, are closely related to ecosystem functioning and biophysical processes [4], [5] and serve as critical indicators for assessing plant condition and stress responses [6], [7]. Therefore, accurate characterization of vegetation canopy conditions is essential for understanding plant physiology, evaluating ecosystem dynamics, and supporting environmental and agricultural applications [8].

With recent advances in remote sensing technologies, uncrewed aerial vehicles (UAVs) equipped with hyperspectral sensors have become an effective platform for high-resolution land-surface and vegetation observation [9], [10]. UAV-based hyperspectral imagery with high spatial and spectral resolution can provide detailed characterization of canopy structure, spectral responses, and spatial features across different environmental conditions [11]. Compared to satellite or crewed airborne platforms, UAV systems have advantages such as operational flexibility, high temporal resolution, and low deployment costs [12]. These advantages provide the capacity for multitemporal and multiscale observations through adjustable flight altitudes and sensor configurations, supporting task-specific monitoring [13], [14]. Accordingly, UAV hyperspectral data has become an important data source for retrieving vegetation biochemical and structural information and for supporting high-resolution remote sensing analysis in complex terrestrial environments [15].

However, despite the numerous advantages of UAV-based hyperspectral remote sensing, the acquired imagery remains susceptible to different illumination conditions, among which shadowing is one of the most prevalent and challenging issues [16]. At high spatial resolutions, the height and 3-D structure of vegetation often cause occlusion under illumination, thereby generating shadows [17]. These shadows can significantly reduce the reflectance of pixels and change the true spectral

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and spatial characteristics of vegetation, which may affect subsequent tasks such as image classification, segmentation, and the retrieval of vegetation biophysical parameters [18], [19]. Even advanced machine learning or deep learning methods may still struggle to fully eliminate these effects if shadow influences are not explicitly addressed [20], [21]. Moreover, shadows may obscure subtle spectral variations associated with plant diseases, stress, or nutrient deficiencies, leading to inaccurate quantitative interpretation of hyperspectral observations [22]. Although the characteristics are weakened in shadowed regions, they still contain recoverable information. Therefore, shadow removal or compensation is commonly adopted as a preprocessing step to improve the reliability of high-resolution hyperspectral observations.

For hyperspectral imagery, there are a variety of approaches for shadow removal, including the use of auxiliary data sources combined with physical modeling. LiDAR data combined with physically based models have been used to correct hyperspectral features and compensate for shadow effects [23]. Polarization-based techniques, which utilize images acquired with and without polarizing filters to estimate the contributions of direct and diffuse radiation, have also been employed for shadow correction [24]. In addition, multispectral data have been integrated to jointly address shadow removal and noise reduction in hyperspectral imagery [25]. External digital surface models (DSMs) combined with radiative transfer models have been used to estimate surface reflectance in shadowed regions [26]. While these approaches can achieve reliable correction under well-constrained conditions, their dependence on additional data sources limits their applicability and operational flexibility, particularly in UAV-based scenarios.

In addition to external data-driven approaches, there are methods using the spectral characteristics inherent in the imagery itself to infer relationships between sunlit and shadowed reflectance. Traditional image-based techniques for shadow removal include the linear correlation correction (LCC) method [27] and histogram matching [28]. More recently, deep learning-based approaches have also been applied to hyperspectral shadow removal [29], [30]. Although these methods show learning capability, their dependence on large training datasets and limited physical interpretability constrain their applicability to high-resolution UAV imagery.

In recent years, spectral unmixing-based approaches have attracted growing interest for shadow removal in hyperspectral imagery, due to their ability to make use of rich spectral information [31]. The shadow multilinear mixing (SMLM) model estimates the degree of shadowing by modeling the interactions among endmembers [32]. Building on this work, extended formulations incorporating multiple illumination sources have been proposed to improve modeling accuracy [33]. To address overcompensation and undercompensation issues in vegetation canopy regions, a triple SMLM (triple-SMLM) strategy introduced a three-stage unmixing framework for near-ground hyperspectral imagery [34]. Spectral angle distance (SAD) has been employed to establish correspondence between endmembers in sunlit and shadowed regions with investigations of nonlinear unmixing algorithms for shadow correction [35]. Spatial information has also been incorporated by jointly modeling diffuse solar illumination and secondary scattering from adjacent pixels [36]. Furthermore, bilinear mixing mechanisms combined with the strong sparsity of shadow distributions have been used to estimate material abundances inside and outside shadowed regions, resulting in improved spectral restoration performance [37]. Despite these efforts, most existing methods either focus on spectral information or incorporate spatial information with fixed regularization strategies, which limits their flexibility in handling complex shadow conditions.

Nevertheless, several challenges remain in shadow removal. Most existing methods only focus on the spectral characteristics of pixels without considering spatial neighborhood information, leading to boundary discontinuities and noise amplification in shadow regions. While some approaches incorporate spatial features to enhance structural consistency, they typically rely on fixed or empirically determined weights to balance spectral and spatial information. However, shadow characteristics vary with spatial resolution, leading to scale-dependent spectral distortion and variations in boundary characteristics. This makes fixed regularization strategies insufficient for achieving a consistent spectral–spatial balance. As a result, existing methods often exhibit inconsistent performance across different spatial scales. Therefore, an adaptive mechanism is required to adjust spatial constraints according to scale-dependent shadow characteristics.

To address the above limitations, we propose a spectral–spatial adaptive weighted method (SAWM) for shadow removal in vegetation canopies, specifically designed to accommodate multiscale UAV-based hyperspectral imagery. The main contributions of this work are summarized as follows.

1) A spectral–spatial joint optimization framework with TV-based regularization is developed to address boundary discontinuities and noise amplification in shadow regions, improving spectral fidelity while preserving spatial consistency.

2) An adaptive weighting strategy based on reconstruction error (RE) and mean spectral deviation (MSD) is proposed to automatically determine the regularization weight, enabling the model to handle scale-dependent shadow-induced spectral variations.

3) The proposed SAWM is evaluated using multiscale UAV hyperspectral datasets over representative vegetation canopies. Experimental results demonstrate consistent improvements over existing methods across different spatial resolutions, providing a reliable spectral basis for subsequent vegetation analysis.

II. Study Area and Data

A. Study Area and Data Collection

The study area is located within an agricultural field operated by Baihua Petroleum Technology Company Ltd., in Shengyuan Street, Dongying District, Dongying, Shandong Province, China (37°25'11.88" N, 118°26'33.87" E, see Fig. 1). It has a warm temperate monsoon continental

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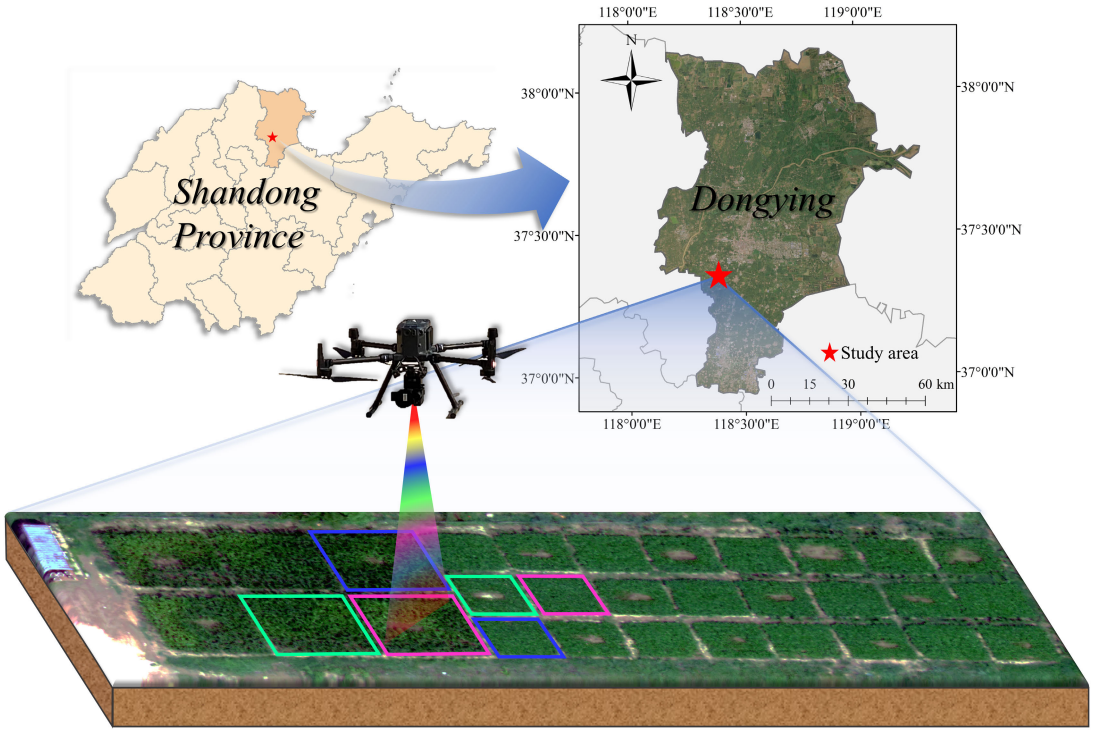


Fig. 1. Study area and experimental plots.

climate, characterized by hot summers, cold winters, and distinct seasonal variations. The average annual temperature is 12.8 °C, and the annual precipitation is approximately 586 mm, providing favorable conditions for vegetation growth. Three types of vegetation were planted in the experimental plots: soybean, maize, and wheat. The soybean plots measured 5×5 m, while the maize and wheat plots measured 8×8 m. Soybean and maize were sown on June 4, 2024, and wheat was sown on October 5, 2023. Uniform field management, including irrigation and weeding, was maintained throughout the growth period.

Hyperspectral data were acquired using a DJI Matrice 300 RTK UAV equipped with a Cubert Ultris X20 Plus hyperspectral camera. Data acquisition was conducted in clear weather around local solar noon. For both soybean and maize, UAV-based hyperspectral images were collected on August 8, 2024 at four spatial resolutions: 1, 2, 4, and 8 cm. For wheat, the images were collected on May 1, 2024 at four spatial resolutions: 1, 2, 4, and 7 cm. For each vegetation, all spatial resolutions were collected within approximately 1 h during a single flight campaign, with flight altitude gradually increased from low to high. This acquisition strategy ensured that changes in spatial resolution primarily reflected variations in ground sampling distance while minimizing the influence of illumination changes. The camera covers a spectral range of 350–1002 nm, with 4-nm spectral intervals.

B. Image Preprocessing

The raw UAV-acquired hyperspectral data were subjected to radiometric correction, image mosaicking, and geometric correction. These preprocessing steps were performed using the official software packages Cubert Utils Touch and Agisoft Metashape. Subsequently, georeferencing was performed to ensure spatial alignment of the hyperspectral imagery. The formula for radiometric correction is shown in the following equation:

$$I_{Trg} = \frac{DN_{Trg} - DN_{Dark}}{DN_{White} - DN_{Dark}} \tag{1}$$

where DN_{Trg} is the raw digital number (DN) of the target pixel, DN_{Dark} is the dark reference used to remove sensor noise, DN_{White} is the white reference DN, and I_{Trg} denotes the radiometrically corrected reflectance of the target.

UAV-based hyperspectral imagery often contains noise that may compromise subsequent data analysis. To mitigate this issue, Savitzky–Golay filtering was applied to smooth the spectral curves and suppress noise. In addition, to avoid potential instability of the sensor at the beginning and end of acquisition, the spectral bands at both ends were removed. The final spectral range used in this study was 400–1000 nm. Image cropping was also performed to ensure that only vegetation and soil were included in the analysis.

III. Methods

A. Shadow Feature Analysis Under Multiscale

The spectral and spatial characteristics of shadow regions are different in various scales of hyperspectral data, which directly affects the accuracy and applicability of shadow removal methods. To investigate this issue, shadow feature analysis under multiscale was performed as part of the preanalysis of shadow removal. Representative spectral and spatial attributes of shadowed regions were extracted and quantified

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using UAV hyperspectral datasets acquired at four spatial resolutions.

From a spectral perspective, sunlit and shadowed regions were distinguished using a sunlit region detection algorithm. The average spectral reflectance curves were then calculated for each region to characterize their overall spectral differences. To further quantify the degree of spectral variation, two metrics were employed.

1) *Spectral Root-Mean-Square Error (SRMSE):* It can measure the absolute difference between the average spectra of the sunlit and shadowed regions across all wavelengths.

2) *Spectral Angle Mapper (SAM):* It calculates the angle between spectral vectors to evaluate the similarity in spectral shape.

The following equations provide the formulas for the two metrics:

$$\text{SRMSE} = \sqrt{\sum_{i=1}^{d} \left(\bar{R}_i^{sunlit} - \bar{R}_i^{shadowed}\right)^2} \tag{2}$$

$$\text{SAM} = \cos^{-1}\left(\frac{\sum_{i=1}^{d} \bar{R}_i^{sunlit} \cdot \bar{R}_i^{shadowed}}{\sqrt{\sum_{i=1}^{d} \left(\bar{R}_i^{sunlit}\right)^2} \cdot \sqrt{\sum_{i=1}^{d} \left(\bar{R}_i^{shadowed}\right)^2}}\right) \tag{3}$$

where $\bar{R}_i^{sunlit}$ is the average reflectance of the sunlit region at band i, $\bar{R}_i^{shadowed}$ is the average reflectance of the shadowed region at band i, and d is the total number of spectral bands.

From a spatial perspective, two metrics were extracted to describe the structural distribution of shadowed regions across different spatial resolutions: the shadow area and the mean shadow edge gradient magnitude (MSEGM). The MSEGM can be used to analyze brightness variation along shadow boundaries. It was calculated by applying a sliding window along the direction of shadow edge to measure the local gradient of pixel intensity decreases. The formula is shown in the following equations:

$$G(x,y) = \sqrt{G_x^2(x,y) + G_y^2(x,y)} \tag{4}$$

$$\text{MSEGM} = \frac{1}{N} \sum_{(x,y)\in\Omega_{edge}} G(x,y) \tag{5}$$

where $G_x^2(x,y)$ and $G_y^2(x,y)$ are horizontal and vertical gradients calculated by the Sobel operator, $G(x,y)$ is the gradient magnitude at pixel, Ω_{edge} is the set of pixels located on the shadow boundary, and N is the total number of pixels in the shadow boundary region.

B. Proposed Shadow Removal Method: Spectral–SAWM

The flowchart of SAWM is shown in Fig. 2. It consists of three main steps. First, sunlit regions were preliminarily detected using a combination of spectral indices and threshold-based segmentation. Second, sunlit endmembers were extracted by unsupervised methods within the sunlit regions and then merged using K-means clustering to reduce redundancy. Finally, shadow removal was conducted using a joint optimization framework that integrated the SMLM with a total variation (TV) regularization term, in which the regularization weight was dynamically adjusted using two indicators.

1) Preliminary Sunlit Region Detection: Prior to shadow removal, it is important to make a preliminary distinction between sunlit and shadowed regions to facilitate the subsequent extraction of endpoints that are not affected by shadows [38]. The natural gas stress index (NGSI) shows distinct value distributions for sunlit soil pixels relative to other regions, which facilitates their identification [39]. Similarly, the transformed chlorophyll absorption in reflectance index (TCARI) shows significant differences between sunlit vegetation and other pixels, making it suitable for extracting sunlit vegetation regions. The following equations present the formulations of these two indices:

$$\text{NGSI} = \frac{\rho_{645} + \rho_{690}}{\rho_{800}} \tag{6}$$

$$\text{TCARI} = 3 \times \left[(\rho_{700} - \rho_{670}) - 0.2 \times (\rho_{700} - \rho_{550}) \cdot \frac{\rho_{700}}{\rho_{670}}\right]. \tag{7}$$

In this study, NGSI and TCARI were used to extract sunlit soil and sunlit vegetation, respectively. NGSI-based extraction employed an empirical threshold, while TCARI-based extraction utilized Otsu's thresholding method [34]. The two resulting regions were then merged to form the overall sunlit region.

2) Sunlit Endmember Extraction: Endmember extraction in the detected sunlit region is a key step for subsequent spectral unmixing and shadow removal. In high-resolution images, there are often mixed pixels at the boundaries of land features, which show overlapping characteristics in the spectrum. If these pixels are directly used in endmember extraction, the resulting endmembers may become contaminated, thereby reducing the accuracy of spectral unmixing. To ensure spectral purity, the Canny edge detector [40] was applied to exclude pixels with strong edge features from the sunlit region before endmember extraction.

Subsequently, the classic unsupervised vertex component analysis (VCA) [41] was employed to extract endmembers. VCA is computationally efficient and requires no manual intervention, making it suitable for large amounts of UAV hyperspectral data processing. To determine the number of endmembers required by VCA, the hyperspectral signal subspace identification by minimum error (Hysime) algorithm was employed [42]. Based on a minimum RE criterion, Hysime can provide a robust and objective estimation of the endmember count, avoiding uncertainty from arbitrary manual selection.

Due to subtle differences in illumination conditions or surface properties among the same type of land cover in natural scenes, the directly extracted endmembers may contain redundant spectra with similar characteristics. To address this issue, the K-means clustering algorithm was used to group similar endmembers based on Euclidean distance, followed by merging of redundant ones [33], [43]. The number of clusters was determined using the Elbow method [44], which identifies the optimal balance between clustering performance

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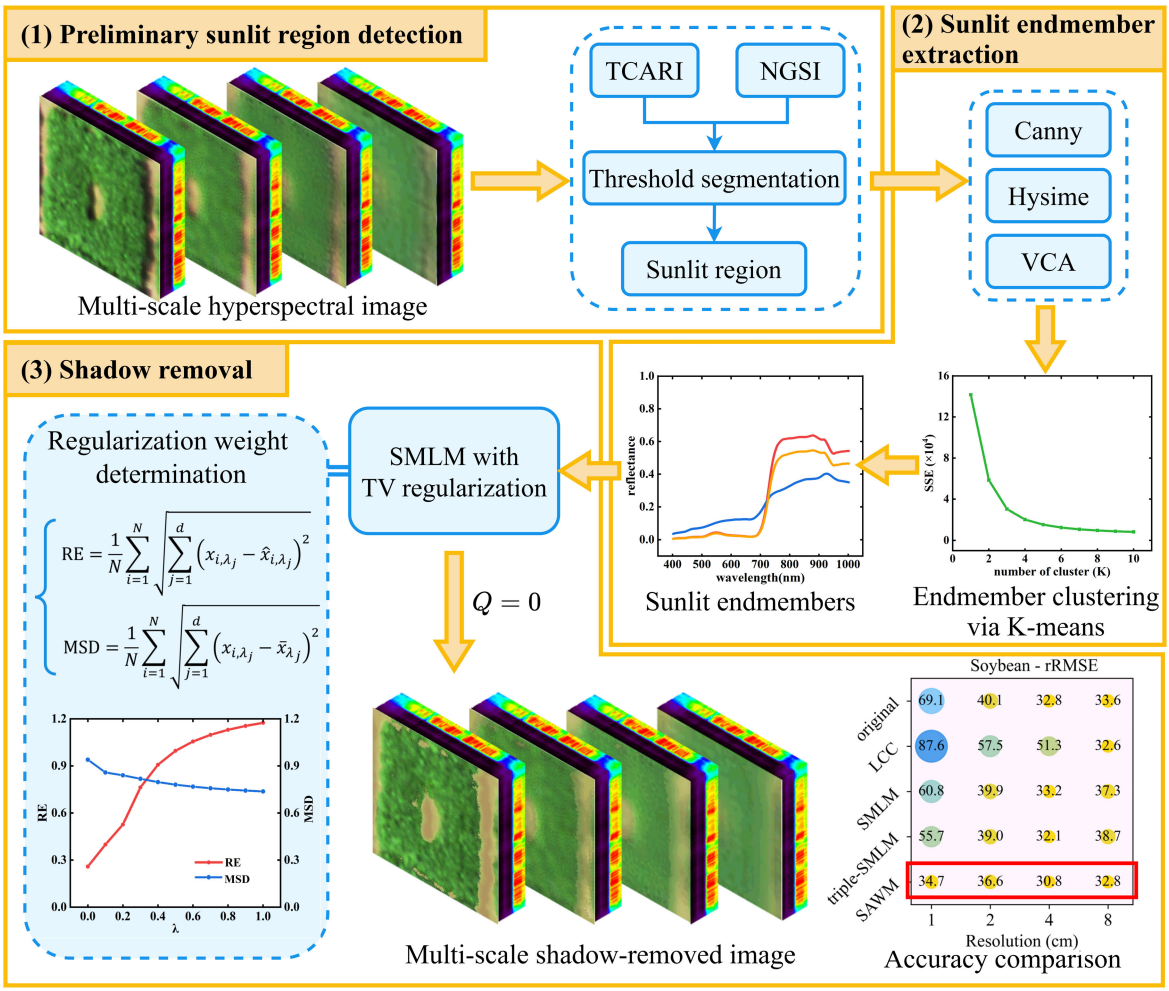


Fig. 2. Flowchart of the spectral–SAWM.

and model complexity based on the within-cluster sum of squares (WCSS) curve. This process ensures that the final endmember set maintains both spectral distinctiveness and representativeness.

Although this procedure includes multiple steps, most components are parameter-free or employ automatically estimated values. The Canny edge detector adopts its default adaptive thresholding scheme and is applied to the first spectral band. Hysime automatically estimates the signal subspace, and the number of endmembers for VCA is derived accordingly [45]. For K-means clustering, the number of clusters is evaluated over a predefined range (1–10) using the Elbow method [46]. The optimal cluster number is selected at the inflection point of the WCSS curve for each dataset. Overall, the workflow requires minimal manual tuning, ensuring stability and reproducibility.

3) Shadow Removal: It is given as follows.

a) Unmixing algorithm with regularization: Building on the bilinear mixing model, the multilinear mixing (MLM) model was developed by extending the interactions among endmembers to an arbitrary order [47]. This model assumes that each light ray interacts with at least one endmember, and after each interaction, it has a probability P of continuing to interact with another endmember. Correspondingly, there is a probability $(1 - P)$ that the light ray reaches the observer directly. The SMLM model represents a further extension of the MLM [32]. A light ray in this model can result in three outcomes following its initial interaction: 1) direct observation; 2) secondary interaction with another endmember; or 3) absorption when the observer is occluded. The probability of the third outcome is defined as Q, which represents the degree of shadowing. The mixing formula of the model is presented in the following equation:

$$\hat{\boldsymbol{x}} = \frac{(1-P)\sum_{i=1}^{p} a_i \boldsymbol{e}_i}{1 - P\sum_{i=1}^{p} a_i \boldsymbol{e}_i} - Q(1-P)\sum_{i=1}^{p} a_i \boldsymbol{e}_i \tag{8}$$

where $\hat{\boldsymbol{x}}$ is the reconstructed reflectance vector of the image pixel, $\boldsymbol{e}_i$ denotes the endmember with abundance a_i, the number of endmembers is $p, a_i \geq 0, \sum_{i=1}^{p} a_i = 1$, and $0 \leq P \leq 1$.

However, the SMLM model only considers spectral information and treats each pixel independently. To enhance the robustness of the model to spatial structure, a TV regularization term is introduced to enforce spatial smoothness while preserving edge features in the reconstruction. The regularization term is defined as shown in the following

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equation:

$$\mathcal{R}_{\mathrm{TV}} = \sum_{k\in\mathcal{N}_j}\sum_{i=1}^{p}|a_{ij}-a_{ik}| \tag{9}$$

where $\mathcal{R}_{\mathrm{TV}}$ represents the TV regularizer, a_{ij} represents the abundance vector of the pixels, and $\mathcal{N}_j$ denotes the set of neighboring pixels around pixel j.

The final objective is to minimize a joint cost function that combines the spectral RE derived from (8) and the regularization term defined in (9). By adjusting the weighting parameter, the model achieves adaptive regularization across different spatial scales. The resulting nonlinear unmixing formulation is presented in the following equation:

$$\min_{a_j}\left\|\boldsymbol{x}_j-\hat{\boldsymbol{x}}_j\left(a_j\right)\right\|^2+\lambda\cdot\mathcal{R}_{\mathrm{TV}}\left(\boldsymbol{a}_j\right) \tag{10}$$

where λ is a regularization parameter to control the influence of spatial constraints, and its value is related to the spatial information of the hyperspectral image.

b) Regularization weight determination based on two-indicator balance points: The optimal regularization parameter λ was to be determined by a quantitative evaluation strategy based on two complementary indicators: the RE and the MSD. These metrics assess the effectiveness of shadow removal from the perspectives of spectral reconstruction capability and global spectral consistency. RE measures the discrepancy between the observed spectrum and the spectrum reconstructed from the estimated model parameters, reflecting the spectral reconstruction capability of the model. MSD quantifies the deviation from the global spectral distribution, representing overall spectral consistency. With changes in spatial resolution, shadow effects exhibit scale-dependent characteristics, such as variations in spectral contrast and boundary mixing, leading to different responses of RE and MSD. Therefore, the balance between these two indicators provides a practical criterion for adaptively determining the regularization parameter λ.

Specifically, the corresponding RE and MSD values were calculated for each candidate value of λ, where λ belongs to the set $\{0, 0.01, 0.02, 0.03, \ldots, 0.1, 0.2, 0.3, \ldots, 1\}$. The formulas for these two indicators are given in the following equations:

$$\mathrm{RE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\sum_{j=1}^{d}\left(x_{i,b_j}-\hat{x}_{i,b_j}\right)^2} \tag{11}$$

$$\mathrm{MSD} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\sum_{j=1}^{d}\left(x_{i,b_j}-\bar{x}_{b_j}\right)^2} \tag{12}$$

where x_{i,λ_j} and $\hat{x}_{i,\lambda_j}$ are the reflectance of the pixel i at band b_j, $\bar{x}_{i,b_j}$ is the mean reflectance of the image at band b_j, N is the number of pixels, and d is the number of spectral bands.

The RE and MSD values are plotted as functions of λ, and the optimal regularization strength is determined by identifying the point where the two curves are closest to each other. Formally, the optimal λ^* is defined in (13). RE and MSD share the same mathematical formulation and are computed using the same spectral distance metric. As a result, they exhibit consistent scaling behavior and are directly comparable. Therefore, no additional normalization is required when using their difference to determine the regularization parameter λ

$$\lambda^* = \arg\min_{\lambda}\left|\mathrm{RE}\left(\lambda\right)-\mathrm{MSD}\left(\lambda\right)\right| \tag{13}$$

where $\mathrm{RE}(\lambda)$ and $\mathrm{MSD}(\lambda)$ represent the spectral RE and the mean spectral distance under λ.

C. Precision Assessment

To assess the difference between the shadow-removed spectra and the reference spectra, two commonly used metrics are adopted: root-mean-square error (RMSE) and relative RMSE (rRMSE). These metrics provide a comprehensive assessment of the shadow removal performance in terms of absolute and relative errors.

Since the true reflectance of shadowed regions under direct illumination is unavailable, the reference spectrum x_{ref} is approximated using the reflectance values of neighboring sunlit pixels with similar surface properties. For the original image, RMSE and rRMSE are calculated by directly comparing the reflectance of shadowed pixels with that of neighboring sunlit pixels. The formulas for RMSE and rRMSE are provided in the following equations:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}\sum_{j=1}^{d}\left(x_{ref_i,\lambda_j}-x_{shd_i,\lambda_j}\right)^2}{N\cdot d}} \tag{14}$$

$$\mathrm{rRMSE} = \sqrt{\frac{1}{N\cdot d}\sum_{i=1}^{N}\sum_{j=1}^{d}\left(\frac{x_{ref_i,\lambda_j}-x_{shd_i,\lambda_j}}{x_{ref_i,\lambda_j}}\right)^2}\times 100\% \tag{15}$$

where x_{shd} is the reflectance of the pixel in the original shadowed region from the shadow-removed image, x_{ref} denotes the reference reflectance approximated from neighboring sunlit pixels with similar surface properties, N is the number of selected pixels, and d is the number of spectral bands. The selection of neighboring sunlit reference pixels followed the boundary-based evaluation principle described in [33]. The specific selection process is detailed in Fig. S1 in the Supplementary Materials.

IV. Results

A. Shadow Feature Under Multiscale

As a preanalysis for shadow removal, a multiscale analysis was conducted to characterize the spectral and spatial properties of shadowed regions. For each vegetation type, all analyses were performed on the same experimental plot imaged at different spatial resolutions. Sunlit and shadowed regions were consistently identified using the same detection strategy described in Section III-B. Mean spectral curves were obtained by averaging pixel-level reflectance within each region, and spectral differences were quantified using SRMSE and SAM. Spatial characteristics were described using shadow area and MSEGM at each spatial resolution.

Fig. 3 shows the mean spectral curves of sunlit and shadowed regions for the three vegetation types at different spatial resolutions. For the sunlit regions, some spectral variations

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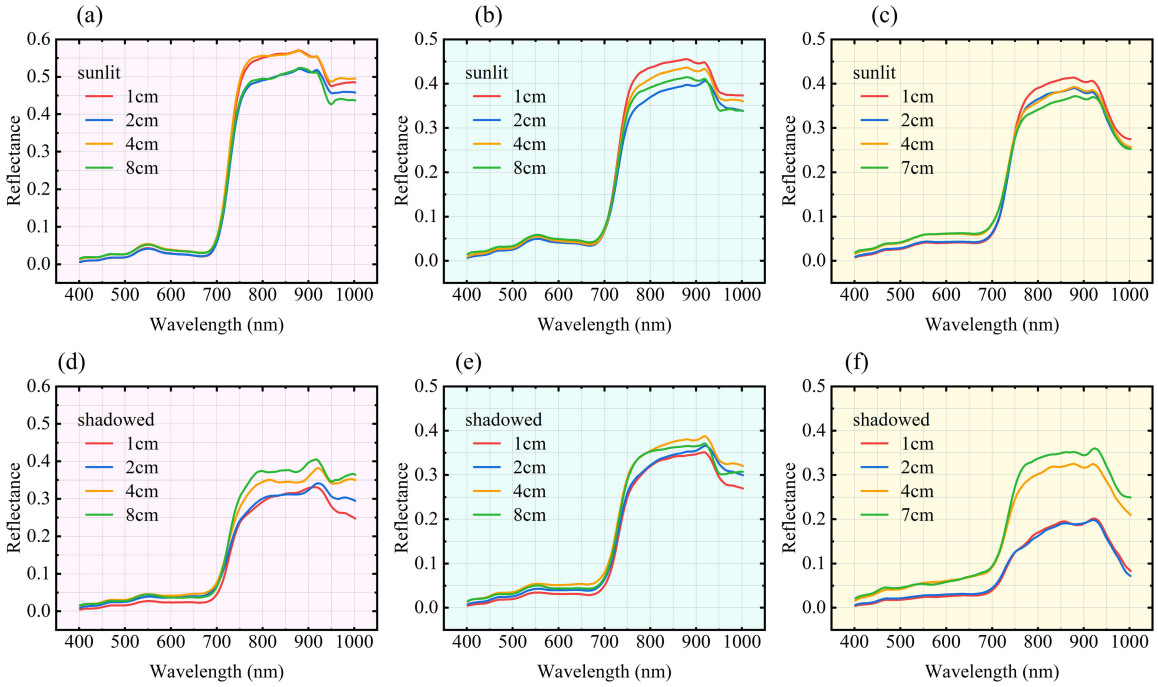


Fig. 3. Mean spectral curves of sunlit and shadowed regions for soybean, maize, and wheat at different spatial resolutions. (a)–(c) Sunlit regions and (d)–(f) shadowed regions.

were observed among different vegetation as spatial resolution changed. In soybean and maize plots, the average reflectance exhibited a nonmonotonic pattern. It decreased from 1 to 2 cm, slightly increased at 4 cm, and then declined again at 8 cm before showing a modest rebound. This trend was particularly pronounced in the near-infrared region [Fig. 3(a) and (b)]. These variations suggested that the canopy structures of soybean and maize were more sensitive to spatial mixing effects. The textures of high-resolution images became blurred at lower resolution and mixed with brighter neighboring areas, causing nonlinear changes in reflectance. In contrast, the images of wheat showed a more consistent spectral response, with the average reflectance in sunlit regions gradually decreasing from 1 to 7 cm [Fig. 3(c)]. This indicated that the wheat canopy exhibited a more uniform spectral distribution, making it less susceptible to pixel aggregation effects.

Compared to the diverse trends observed in sunlit regions, the spectral response of shadowed regions showed a more consistent pattern across all three types of vegetation. For soybean, maize, and wheat, the average reflectance in shadowed regions gradually increased as the spatial resolution decreased [Fig. 3(d)–(f)]. This trend was primarily attributed to the rising proportion of mixed pixels at lower resolutions. At lower spatial resolution, a single pixel was more likely to include both shadowed and partially sunlit vegetation, which elevated the overall reflectance in shadowed regions. Additionally, lower resolution tended to smooth local variability within shadows, further increasing the observed reflectance. Although shadows appeared visually dark, their spectral reflectance could rise as spatial resolution became lower.

To quantify the spectral differences between sunlit and shadowed regions, the mean spectral curves of the two regions were compared, and two commonly used metrics were adopted for evaluation: SRMSE and SAM. For all three vegetation types, SRMSE generally decreased as the spatial resolution became lower [Fig. 4(a)–(c)]. A slight increase was observed between 2 and 4 cm for soybean and maize, though the overall trend remained decreasing. This fluctuation was caused by increased spectral mixing and blurred boundaries, which complicated the separation of sunlit and shadowed regions. In contrast, SAM showed no clear consistent monotonic trend across resolutions, possibly due to spectral shape variability over the wavelength range.

To characterize the spatial distribution of shadowed regions, the shadow area and MSEGM were calculated at each resolution. As shown in Fig. 4(d)–(f), the shadow area decreased with lower spatial resolution across all three vegetation types. This trend was likely related to the blurring of the shadowed boundaries and the averaging of small shadowed patches at lower resolutions. In contrast, MSEGM steadily increased as the resolution became lower, reflecting sharper brightness variation along shadow boundaries. With larger pixel sizes, shadowed and nonshadowed regions were more likely to be mixed within the same pixel, which enhanced the intensity difference at shadow boundaries. Since gradient operators were sensitive to local brightness variations, pixels along shadow boundaries exhibited stronger gradient values at lower spatial resolutions, leading to an overall increase in MSEGM.

In summary, with decreasing spatial resolution, the mean reflectance in shadowed regions increased, the absolute

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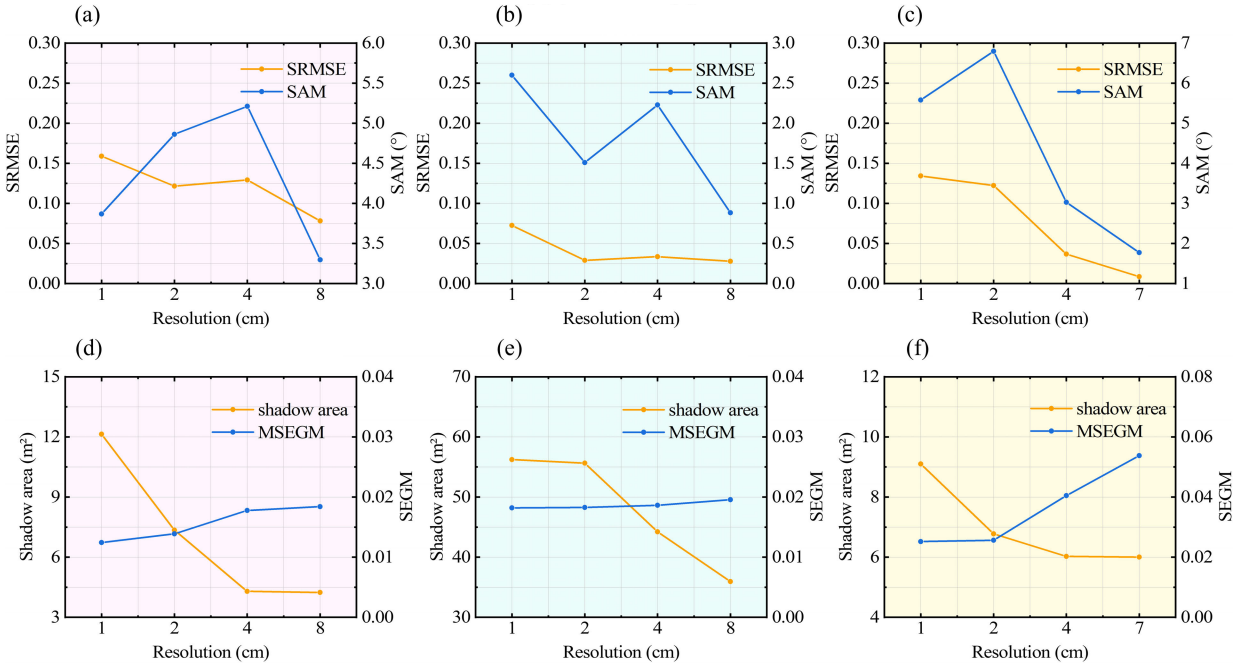


Fig. 4. Spectral and spatial features of soybean, maize, and wheat at different spatial resolutions. (a)–(c) Spectral features measured by SRMSE and SAM. (d)–(f) Spatial features quantified by shadow area and MSEGM.

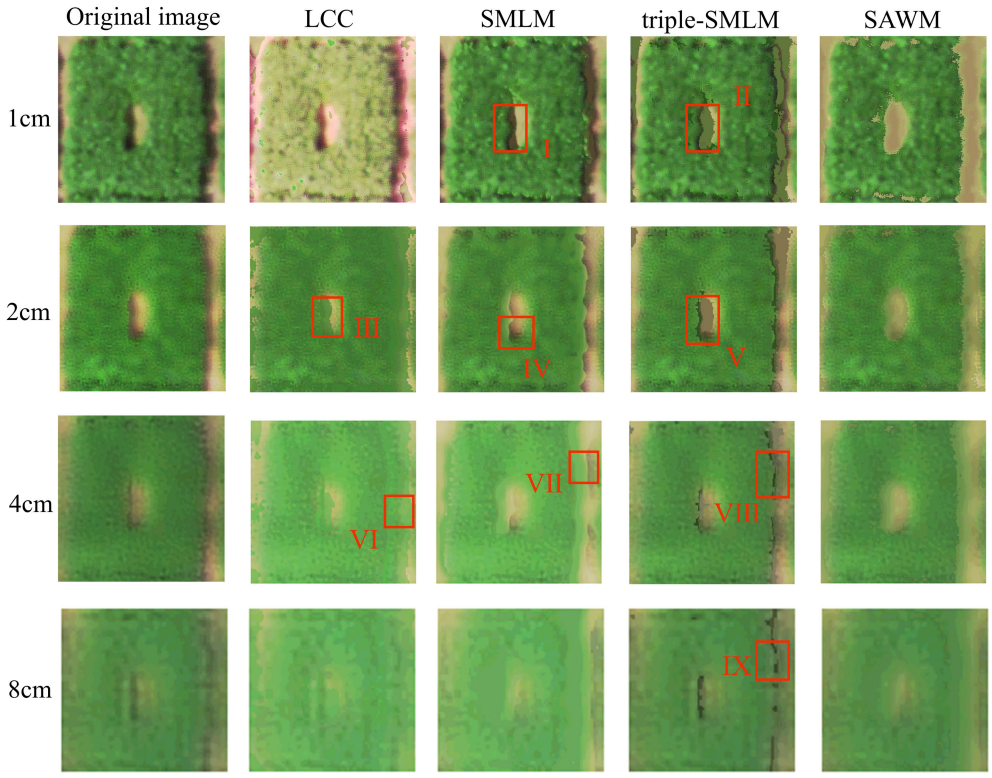


Fig. 5. Comparison of original and shadow-removed images of soybean plots obtained using different shadow removal methods. (Left to right) original image, LCC, SMLM, triple-SMLM, and SAWM. (Top to bottom) 1, 2, 4, and 8 cm.

spectral differences between sunlit and shadowed regions were reduced, the shadow area became smaller, and brightness variation along shadow boundaries increased. These observations provide a basis for analyzing scale-dependent effects in subsequent shadow removal modeling.

B. Performance of Shadow Removal Method Under Multiscale

1) Shadow Removal Results and Visual Assessment: LCC is a representative traditional shadow removal method, while SMLM is an advanced mixing model that performs shadow

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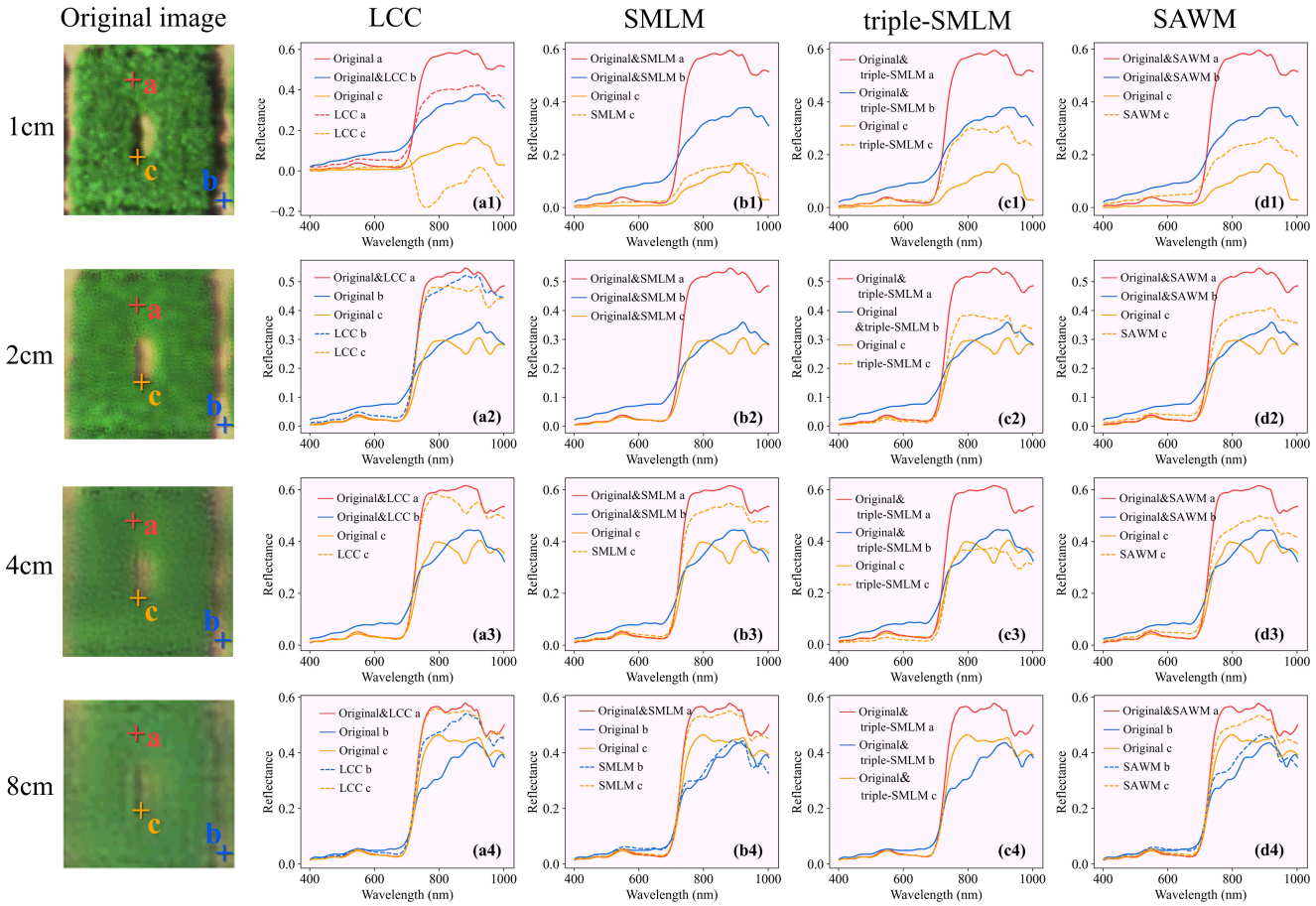


Fig. 6. Spectral comparison between original and shadow-removed results of (a1)–(a4) LCC, (b1)–(b4) SMLM, (c1)–(c4) triple-SMLM, and (d1)–(d4) SAWM for three labeled pixels in soybean plots.

removal by setting $Q = 0$. On this basis, the triple-SMLM, an improved version of SMLM, has been adapted for fine-scale vegetation canopy shadow removal. These three methods were selected for comparison with the proposed SAWM approach. Fig. 5 presents the original and shadow-removed soybean images using different methods.

For soybean, LCC caused overcompensation at the 1-cm resolution, significantly changing the color of sunlit regions. Although the results of LCC at 2 and 4 cm showed some visual improvement, some bare soil regions were incorrectly compensated as vegetation (Fig. 5, regions III and VI). SMLM showed insufficient compensation at 1-cm resolution, with residual shadows remaining in the shadow-removed image (Fig. 5, region I). The results of SMLM at 2 cm showed abrupt transitions, and some shadow regions remain inadequately removed (Fig. 5, region IV). Similar issues were observed at 4 cm, where certain shadow regions remained uncompensated (Fig. 5, region VII). Triple-SMLM showed that the shadow removal produced overly sharp boundaries at 1 and 2 cm (Fig. 5, regions II and V), and failed to remove the subtle shadows at 4 and 8 cm (Fig. 5, regions VIII and IX). In contrast, the proposed SAWM achieved more effective shadow removal across all spatial resolutions. It preserved the original reflectance characteristics of sunlit regions while providing accurate compensation in shadowed regions, including at challenging shadow boundaries. Additional results for other vegetation were shown in Figs. S2 and S3 (in the Supplementary Materials).

Fig. 6 presents the spectral curves of three representative pixels, labeled a, b, and c, selected from soybean plots at different spatial resolutions. Pixels a and b correspond to different sunlit regions, while c represents the shadowed region. In the 1-cm image, LCC changed the original reflectance of the sunlit regions and produced the compensated spectrum in the shadowed region that is inconsistent with the expected values [Fig. 6(a1)]. SMLM provided partial compensation at 1 cm, but the shadow-removed spectrum at point c remained lower than the reference [Fig. 6(b1)]. At 2 cm, SMLM failed to compensate for the shadowed region, as the spectrum was identical to the uncorrected one [Fig. 6(b2)]. The triple-SMLM showed generally acceptable spectral compensation, but in the visible range, the shadow-removed values were lower than the original ones [Fig. 6(c3)], and no processing was applied to small shadow regions such as those at 8 cm [Fig. 6(c4)]. In contrast, SAWM achieved appropriate compensation across all spatial resolutions, restoring the spectral characteristics of shadowed regions without altering those of sunlit regions. Similar results were observed for other vegetation types, as illustrated in Figs. S4 and S5 (in the Supplementary Materials).

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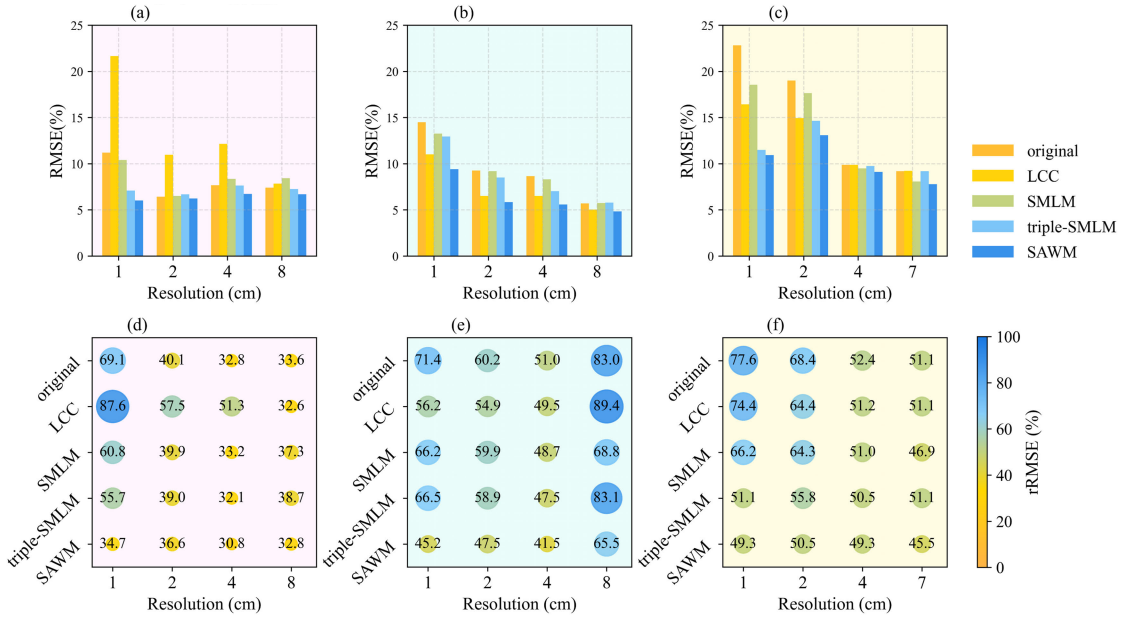


Fig. 7. Comparison of RMSE and rRMSE across three vegetation types for the original and shadow-removed images. (a)–(c) RMSE for soybean, maize, and wheat, respectively. (d)–(f) rRMSE.

TABLE I
RMSE VALUES OF THE ORIGINAL IMAGES AND SEVERAL SHADOW-REMOVED IMAGES USING DIFFERENT METHODS

Type	Resolution	Original	LCC	SMLM	triple-SMLM	SAWM
Soybean	1 cm	0.1119	0.2166	0.1040	0.0709	0.0602
	2 cm	0.0643	0.1096	0.0652	0.0668	0.0623
	4 cm	0.0767	0.1215	0.0835	0.0764	0.0673
	8 cm	0.0741	0.0784	0.0844	0.0725	0.0670
Maize	1 cm	0.1450	0.1102	0.1326	0.1295	0.0931
	2 cm	0.0925	0.0651	0.0920	0.0851	0.0584
	4 cm	0.0865	0.0652	0.0831	0.0702	0.0558
	8 cm	0.0569	0.0503	0.0575	0.0579	0.0484
Wheat	1 cm	0.2282	0.1643	0.1855	0.1149	0.1093
	2 cm	0.1901	0.1494	0.1765	0.1464	0.1309
	4 cm	0.0989	0.0971	0.0949	0.0975	0.0911
	7 cm	0.0919	0.0923	0.0807	0.0919	0.0778
Mean-RMSE		0.1097	0.1100	0.1033	0.0900	0.0768

2) Shadow Removal Accuracy: In this study, RMSE and relative rRMSE were used to assess the accuracy of shadow removal. These two metrics evaluate absolute and relative errors, respectively, across multiple spatial resolutions and vegetation types. Lower RMSE and rRMSE values indicate better shadow removal performance.

Fig. 7(a)–(c) and Table I present the RMSE values of the original images and the shadow-removed results. For soybean images, the RMSE values of LCC were higher than those of the original images across all spatial resolutions, indicating a decrease in spectral accuracy after shadow removal. In contrast, for maize and wheat, LCC slightly improved the results. For SMLM, the RMSE values were lower than those of the original images in all cases except for the soybean image at 2 cm, where the RMSE slightly increased. However, SMLM showed inferior performance compared to LCC in certain cases. For example, in the maize image at 2 cm, the RMSE values were 0.0651 for LCC and 0.0920 for SMLM. Triple-SMLM provided effective shadow removal across different vegetation and spatial resolutions, with a mean RMSE of 0.0900. However, its performance in the 8-cm maize image was limited, where the RMSE was even higher than that of the original image. SAWM consistently outperformed the other methods, with lower RMSE values across all vegetation types and spatial resolutions. The mean RMSE of SAWM across different resolutions and crops was 0.0768, representing an improvement of 0.0330 over the original images. Relative to the other methods, the mean RMSE of SAWM was reduced by 0.0332, 0.0265, and 0.0132 relative to LCC, SMLM, and triple-SMLM, respectively.

Fig. 7(d)–(f) and Table II present the rRMSE values for the original images and the shadow-removed results. For maize and wheat, the rRMSE values of LCC were lower than those of the original images, suggesting a certain degree of shadow removal effectiveness. However, for soybean, the rRMSE values of LCC increased, indicating a reduction in accuracy. These trends are consistent with the RMSE results. SMLM generally improved accuracy across all types of vegetation and resolutions, achieving the best performance in the 4-cm soybean image (rRMSE reached 33.2%). The rRMSE value of triple-SMLM in the 8-cm maize image was higher than that of the original image, showing the same trend as RMSE. As shown in Fig. 7(d)–(f), the rRMSE values of SAWM remained lower than those of the other methods under varying vegetation and spatial conditions, demonstrating superior performance. In addition, SAWM exhibited stable rRMSE values across different spatial scales within each vegetation type, with a

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TABLE II
RRMSE VALUES OF THE ORIGINAL IMAGES AND SEVERAL SHADOW-REMOVED IMAGES USING DIFFERENT METHODS

Type	Resolution	Original	LCC	SMLM	triple-SMLM	SAWM
Soybean	1 cm	69.07%	87.61%	60.82%	55.65%	34.71%
	2 cm	40.07%	57.53%	39.90%	38.97%	36.57%
	4 cm	32.80%	51.33%	33.16%	32.13%	30.83%
	8 cm	33.62%	32.83%	37.31%	38.66%	32.81%
Maize	1 cm	71.38%	56.18%	66.18%	66.50%	45.17%
	2 cm	60.17%	54.91%	59.90%	58.88%	47.47%
	4 cm	50.98%	49.52%	48.65%	47.48%	41.45%
	8 cm	82.99%	89.43%	68.79%	83.13%	65.52%
Wheat	1 cm	77.57%	74.40%	66.17%	51.08%	49.27%
	2 cm	68.37%	64.39%	64.31%	55.83%	50.51%
	4 cm	52.37%	51.22%	50.96%	50.48%	49.33%
	7 cm	51.08%	51.08%	46.94%	51.08%	45.45%
Mean-rRMSE		57.54%	60.04%	53.59%	52.49%	44.09%

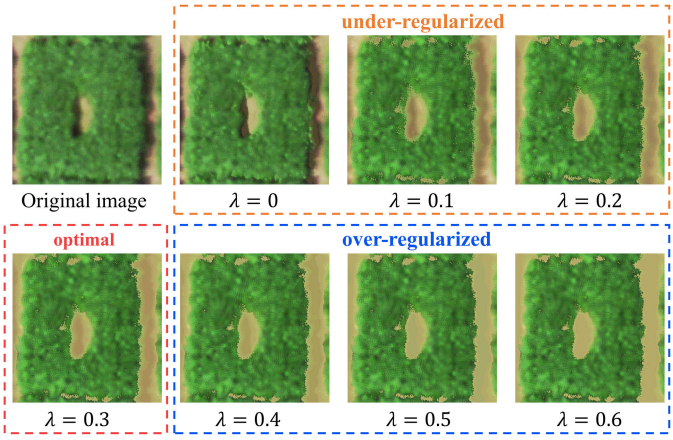


Fig. 8. Visual comparison of shadow removal results under different regularization strengths (under-regularized, optimal, and over-regularized) for the soybean dataset.

mean rRMSE of 44.09%, improving by 13.45% over the original images. In comparison with the other methods, SAWM reduced the mean rRMSE by 15.94%, 9.50%, and 8.40% over LCC, SMLM, and triple-SMLM, respectively.

In summary, SAWM achieved the best performance in both RMSE and rRMSE, indicating its broad applicability under different vegetation and spatial resolution conditions. In addition to the above experiments, further validation was conducted on an independent large-scale aerial hyperspectral dataset acquired over Xiong'an New Area (Matiwan Village) [48]. Detailed results are provided in supplementary material (Figs. S6–S7 and Table S2).

V. DISCUSSION

A. Model Adaptive Performance: Regularization Factor λ

To illustrate the effect of the regularization parameter λ, Fig. 8 presents a visual comparison of shadow removal results under three representative settings, corresponding to under-regularization, optimal regularization, and over-regularization, using the soybean dataset at a spatial resolution of 1 cm. In the under-regularized case, the imposed spatial constraints are insufficient, resulting in incomplete shadow removal, particularly near shadow boundaries. In the optimally regularized case, the reconstructed image achieves a good balance between spectral fidelity and spatial smoothness, leading to effective shadow removal while preserving structural details. In contrast, in the over-regularized case, excessively strong spatial constraints introduce over-smoothing effects, causing loss of spectral details and blurred boundaries. These observations provide an intuitive understanding of the impact of λ.

To further quantify this effect, we evaluated RMSE and rRMSE under different λ values using three types of vegetation data at a spatial resolution of 1 cm (Fig. 9). For soybean, maize, and wheat, both RMSE and rRMSE exhibited a typical nonmonotonic trend. They initially decreased and then increased as λ increased, reaching a minimum at a mid-range value. This trend suggests that the incorporation of the regularization term is essential for improving shadow removal performance. However, excessive regularization may lead to over-smoothing of spectral features, thereby reducing shadow removal accuracy. This reflects a nonlinear adjustment effect. A similar pattern was also reported in [36]. Increasing λ can effectively suppress noise in the abundance map, whereas excessively large values lead to a significant increase in estimation error.

Notably, when λ was set to the optimal value λ^* determined based on two-indicator balance points, both RMSE and rRMSE reached their minimum, outperforming all other tested parameter settings. This finding quantitatively validates the effectiveness of the λ^* selection strategy proposed in this study, further reinforcing the importance of incorporating spatial regularization to enhance shadow removal performance. It also highlights the critical role of parameter tuning in optimizing the performance of the SAWM model.

In addition, the optimal regularization parameter λ^* corresponding to SAWM was computed under different spatial resolutions. As shown in Fig. 10, λ^* generally decreases with decreasing spatial resolution across three vegetation types, with the most obvious trend in soybean. This phenomenon may be closely related to the spatial structural characteristics of the imagery [37]. High-resolution images retain rich texture details and edge information, and stronger regularization constraints need to be applied in order to suppress local noise and enhance spatial consistency. On the other hand, low-resolution images lose structural details due to pixel averaging, where overly strong spatial regularization may lead to oversmoothed compensation. Therefore, the observed variation of the optimal regularization parameter λ^* across different spatial scales reflects the capacity of SAWM to adaptively adjust the strength of spatial constraints.

The analysis in Section IV-A indicates that the MSEGM increases with decreasing resolution, in contrast to the downward trend of λ^*. This increase in MSEGM can be attributed to the sensitivity of gradient operators to brightness discon-

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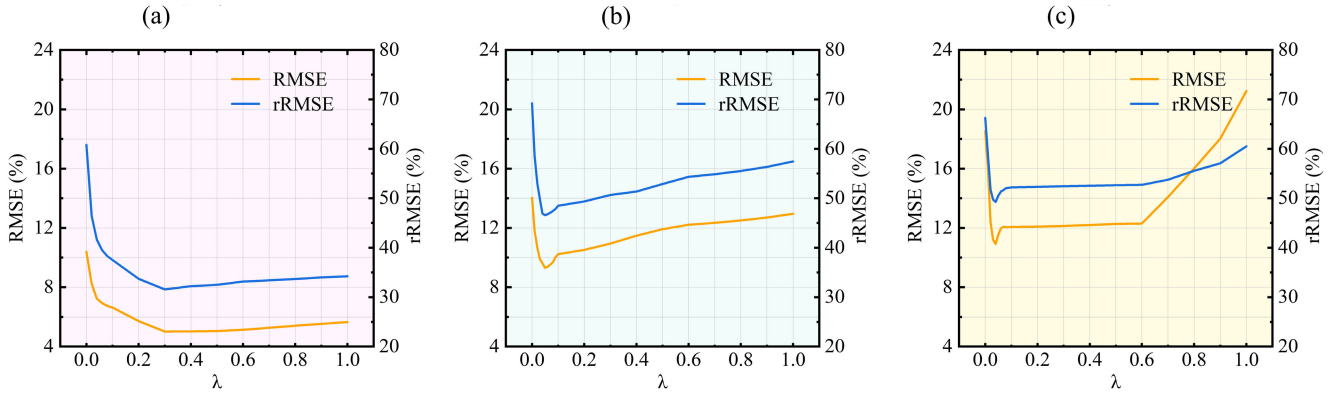


Fig. 9. Effect of regularization parameter λ on RMSE and rRMSE of shadow removal for (a) soybean, (b) maize, and (c) wheat.

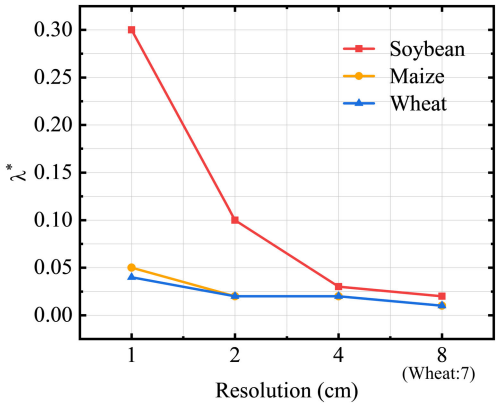


Fig. 10. Variation of the optimal regularization parameter λ^* with spatial resolution in the SAWM.

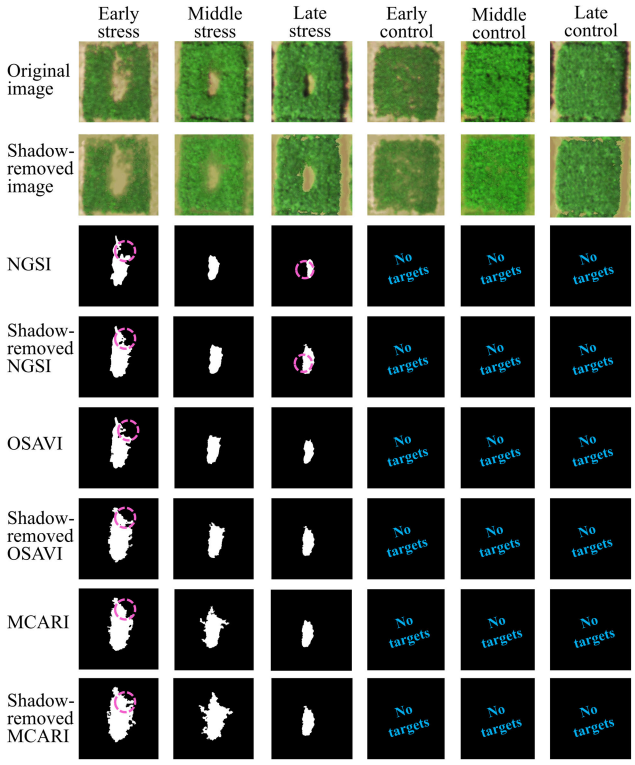


Fig. 11. Results of detecting CO_2-stressed soybean regions using different methods before and after shadow removal.

tinuities, where enhanced edge responses elevate the overall gradient magnitude. Although the rise in MSEGM reflects more visible local intensity variations, the reduction in spatial resolution weakens structural information, limiting the benefits of stronger regularization. Therefore, the decrease of λ^* can be interpreted as a response to diminished spatial detail, highlighting the SAWM algorithm's ability to rationally adjust the strength of regularization under multiscale conditions.

B. Application of the Proposed Shadow Removal Method to Stressed Vegetation Detection

To evaluate the practical applicability of the proposed SAWM in agricultural stress monitoring, it was applied to a vegetation stress detection task under a simulated CO_2 microleakage scenario. This scenario simulates possible CO_2 leakage during underground carbon storage, where physiological stress responses in vegetation are typically subtle and easily masked by shadows in high-resolution UAV-based hyperspectral imagery, making detection particularly challenging. Details of the experimental setup can be found in [49]. Soybean, maize, and wheat were selected as representative vegetation species. To capture early, middle, and late stages of CO_2 leakage stress, hyperspectral images for soybean and maize were collected on July 18, July 31, and August 8, 2024, while wheat images were collected on April 18, May 1, and May 14, 2024.

Three widely used vegetation stress indices were selected to detect stressed regions, including the NGSI [50], the optimized soil-adjusted vegetation index (OSAVI) [51], and the modified chlorophyll absorption in reflectance index (MCARI) [52]. The detection process consisted of three main steps: 1) calculating NGSI, OSAVI, and MCARI indices from the UAV-based hyperspectral imagery; 2) segmenting the images using the Otsu thresholding method; and 3) applying morphological operations to remove fragmented patches and irrelevant elongated features [53].

Figs. 11–13 show the results for several index-based models before and after shadow removal. The experimental results indicate that shadow interference generally leads to incomplete

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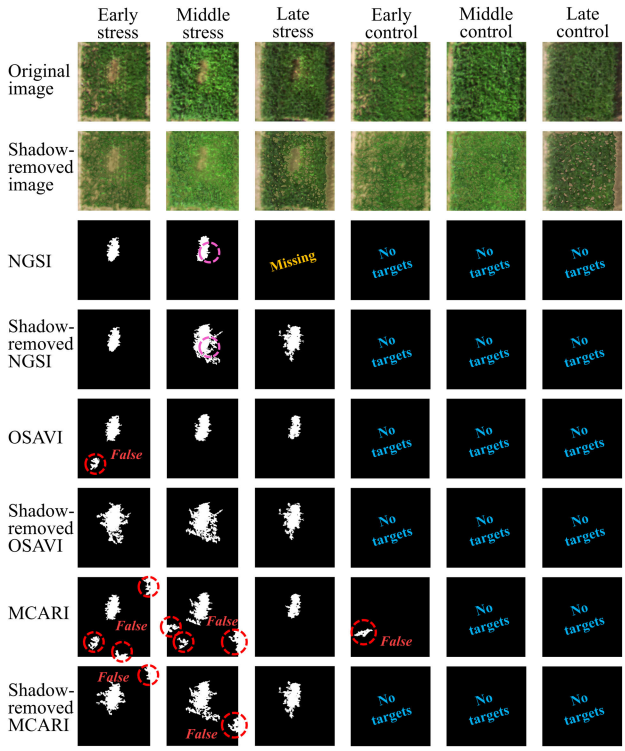


Fig. 12. Results of detecting CO_2-stressed maize regions using different methods before and after shadow removal.

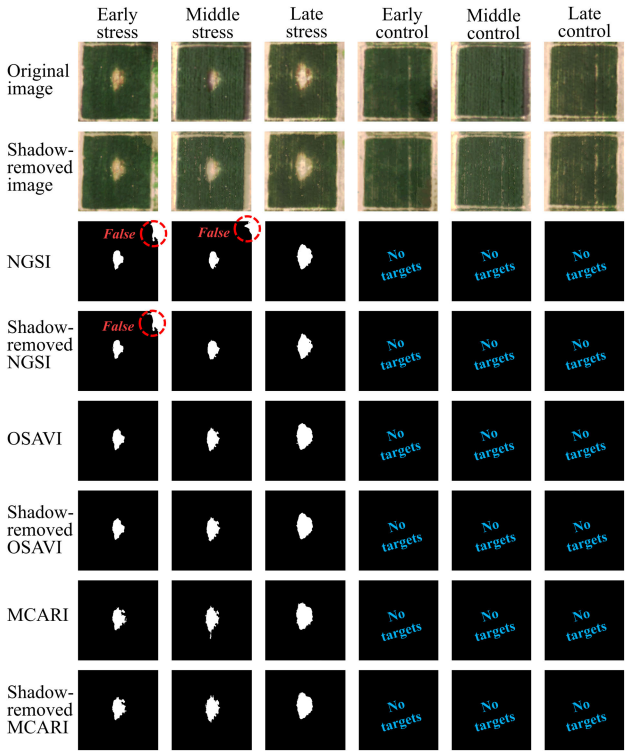


Fig. 13. Results of detecting CO_2-stressed wheat regions using different methods before and after shadow removal.

detection or misdetection of stressed-vegetation areas, especially during early and middle stages. In the maize plots, the taller plants and more complex canopy structures resulted in more severe shadow occlusion. Shadow removal significantly

TABLE III

ACCURACY AND RECALL COMPARISON BEFORE AND AFTER SHADOW REMOVAL FOR THREE VEGETATION TYPES UNDER CO_2 STRESS

Vegetation	Period	Method	Accuracy		Recall	
			Original	Shadow-removed	Original	Shadow-removed
Soybean	Early	NGSI	92.30%	94.80%	52.17%	67.70%
		OSAVI	94.60%	96.10%	66.46%	78.88%
		MCARI	95.90%	96.10%	76.40%	82.61%
	Middle	NGSI	96.10%	96.60%	40.91%	57.50%
		OSAVI	95.60%	96.30%	60.64%	67.75%
		MCARI	94.50%	95.50%	82.50%	87.88%
	Late	NGSI	96.33%	98.00%	42.11%	52.63%
		OSAVI	97.00%	98.00%	52.63%	68.42%
		MCARI	97.67%	98.33%	63.16%	73.68%
Maize	Early	NGSI	88.60%	89.80%	22.14%	23.49%
		OSAVI	89.80%	92.40%	30.53%	53.69%
		MCARI	87.90%	89.80%	28.24%	38.93%
	Middle	NGSI	90.20%	92.80%	23.44%	51.56%
		OSAVI	90.90%	92.40%	28.91%	59.38%
		MCARI	86.10%	89.20%	37.50%	49.22%
	Late	NGSI	91.07%	93.80%	0.00%	50.91%
		OSAVI	94.45%	94.00%	34.15%	52.96%
		MCARI	94.15%	94.80%	35.98%	48.86%
Wheat	Early	NGSI	95.70%	95.52%	43.98%	43.94%
		OSAVI	97.64%	97.68%	51.04%	51.87%
		MCARI	98.18%	98.24%	63.07%	63.90%
	Middle	NGSI	96.32%	97.72%	40.41%	53.47%
		OSAVI	98.06%	98.26%	60.82%	65.31%
		MCARI	98.10%	98.42%	66.94%	72.65%
	Late	NGSI	99.01%	99.02%	80.56%	80.83%
		OSAVI	99.38%	99.46%	92.86%	93.28%
		MCARI	99.16%	99.46%	94.96%	94.96%

reduced the incidence of omission and commission errors in index-based models such as NGSI and OSAVI (Fig. 12). In the soybean plots, there was an improvement in spatial agreement between the detected and the actual CO_2 leakage stress regions after shadow removal (Fig. 11). Despite relatively weak shadow interference in the wheat plots, early stage misidentification by the NGSI model was still reduced to some extent following shadow removal (Fig. 13).

From a quantitative perspective, three vegetation index-based models exhibited improved Accuracy and Recall after shadow removal with SAWM (Fig. 14). For example, in maize plots, the Recall of the NGSI model increased by 28.13% and 50.91% during the middle and late stages, respectively, while the Accuracy of the MCARI model improved by 3.1% during the early stage (Table III). These results indicate that SAWM can provide significant enhancement in stress identification across varying canopy structures and illumination conditions. Moreover, the results are consistent with those reported by [54], further confirming the effectiveness of shadow removal in enhancing stress detection.

Overall, the findings demonstrate that SAWM can not only improve the accuracy of stress identification under complex shadow conditions, but also provide a reliable tool for

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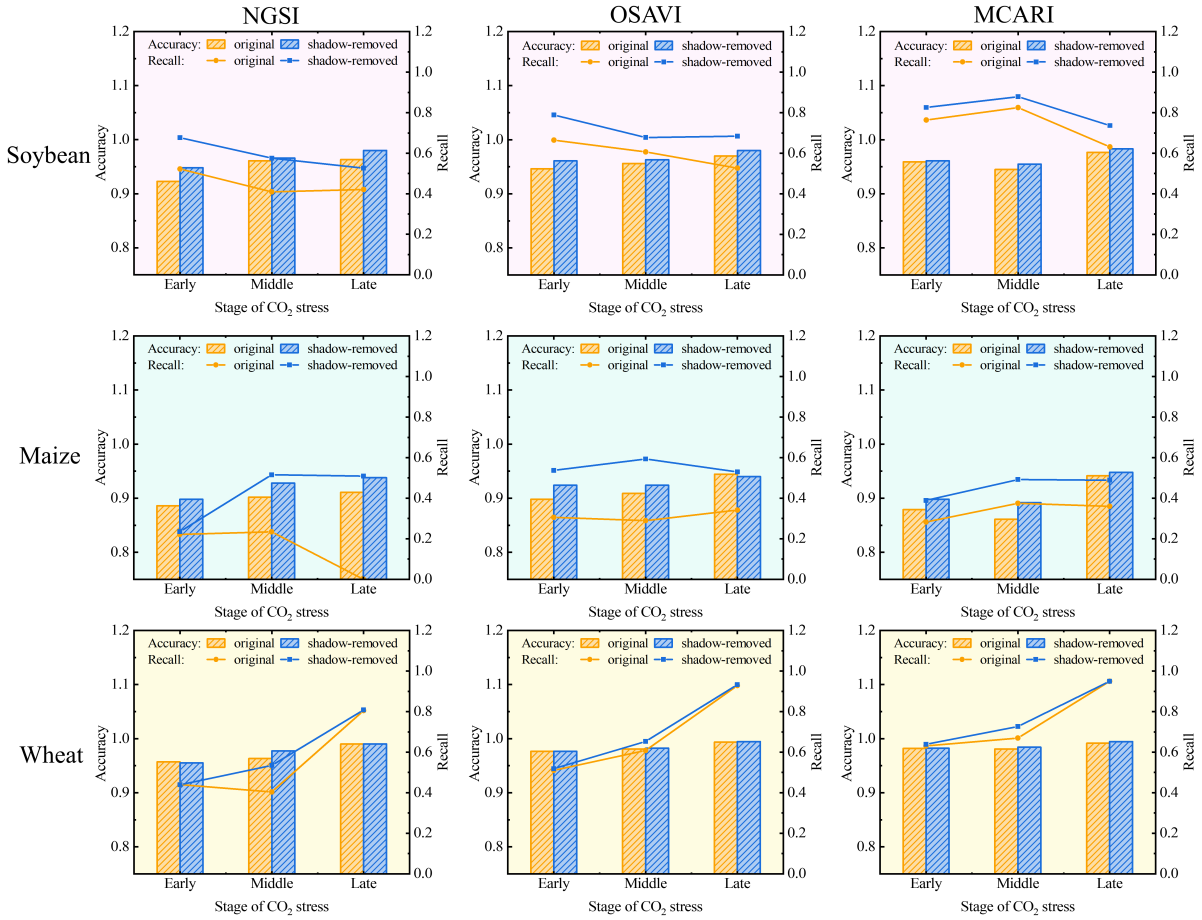


Fig. 14. Evaluation of detection performance for three vegetation types under CO_2 stress before and after shadow removal. (Left to right) NGSI, OSAVI, and MCARI. (Top to bottom) soybean, maize, and wheat.

vegetation stress analysis, showing potential for practical applications in agricultural remote sensing.

VI. Conclusion

This study investigated the spectral and spatial variations of shadowed regions under different spatial resolutions: as the spatial resolution decreased, the mean reflectance in shadowed regions increased, and the absolute spectral differences between sunlit and shadowed regions were reduced. Meanwhile, the shadow area became smaller, and brightness variation along shadow boundaries increased. This suggests that shadow-related spectral and spatial information varies at different spatial scales. Following these analyses, this study proposed an SAWM for shadow removal in multiscale UAV-based hyperspectral imagery of vegetation canopies. Based on spectral unmixing, the method incorporated spatial information through the introduction of TV regularization term to improve spatial continuity at the shadow boundaries. Furthermore, a two-indicator (RE and MSD) balance strategy was developed to adaptively determine the regularization weight, which can enable adjustment within the evaluated multiscale UAV scenarios. The proposed method was evaluated using UAV hyperspectral data of three vegetation types (soybean, maize, and wheat) at four spatial resolutions. The main conclusions are as follows: SAWM achieved a mean RMSE of 0.0768 and a mean rRMSE of 44.09% across all spatial resolutions and vegetation types, demonstrating high accuracy in shadow removal. Compared with the original images, SAWM reduced the mean RMSE and rRMSE by 0.0330% and 13.45%, respectively. Compared with the other methods, SAWM reduced the RMSE by 0.0132–0.0332 and the rRMSE by 8.40%–15.94%, showing consistent improvements across different spatial resolutions.

This study further analyzed the influence of the regularization parameter λ on shadow removal accuracy and examined the variation of the optimal λ^* across different spatial resolutions. The results demonstrate the ability of the proposed SAWM to adaptively adjust to varying spatial scales. Furthermore, the effectiveness of SAWM was assessed in a representative application scenario involving the detection of vegetation under CO_2 microleakage stress, indicating its potential value for precision agriculture and environmental monitoring.

Despite its evaluation across a range of UAV hyperspectral datasets, the proposed strategy still requires further investigation under more complex illumination conditions or broader acquisition settings. Future work will focus on extending the SAWM framework in directions closely related to its core methodology. One important direction is to investigate how complementary structural information, such as LiDAR

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data, can be incorporated into the proposed framework to better characterize shadowed regions. Another direction is to systematically evaluate the transferability of the proposed method across different sensor types, such as from UAV-based imagery to satellite observations. In addition, integrating the current physically based framework with deep learning-based strategies may provide a promising way to enhance the robustness and computational efficiency of shadow removal.

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